Linear Association

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Introduction to Statistics (STAT 1200)

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Topics

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Recap

- In **Hypothesis Testing III** we discussed:
  - Tests for normality.
  - ANOVA.
  - Chi-squared tests for independence.

Introduction

- We are continuing to make statistical inferences about target populations.
- We are interested in the linear association between two or more variables.
  - *Scatterplots*
  - Covariance
  - Correlation
  - Linear Regression

Visualizing Multiple Variables

- After exploring each variable individually, we can assess possible relationships between variables.
- The specific layout will again depend on if your variables are **categorical** or **continuous**.
- **Two continuous variables:**
  - Scatterplot
  - Scatterplot matrix (all continuous variables)
- One categorical and one continuous variable:
  - Faceting (show distributions of each category)
  - Boxplot/violin plot for each category
- Two categorical variables:
  - Cross-tabulation
  - Heatmap

One Variable

- One of the most commonly used R functions for generating visualizations is the `plot()` function.
- In order to generate a scatter plot of a single variable we use `plot(data.name$variable.name)`.
- The `y`-axis will contain the variable values and the `x`-axis will contain the index of the variable (order in the data).
- Many base R visualization functions do not have a `data=` argument.
  - Instead, we can use `with(data.name,plot(variable.name)`

Two Variables

- To generate a scatterplot of two numeric variables we just need to add the `x` and `y` variables as arguments.

- We can use:
  - `plot(x = data.name$variablex.name, y = data.name$variabley.name`
  - `plot(data.name$variabley.name` $\sim$ `data.name$variablex.name)` (common regression notation)

- *Pick which approach works best for you an stick with it.*

- You can also use *ggplot2*.

Preliminary Example 1

- Using the *iris* dataset, create the following scatterplots:
  1. Petal.Length $\sim$ Petal.Width
  2. Sepal.Length $\sim$ Sepal.Width
  3. Petal.Length $\sim$ Sepal.Length
- Do you notice anything in the plots?

Linear Association

Covariance

- The **covariance** is a measure of *linear association* between two random variables.
  - Measures how much and to what extent two variables change together.
  - Measures the direction of the linear association between two random variables.

- Covariance between two (population) random variables X and Y:

$$\sigma_{XY} = Cov(X, Y) = E[(X - \mu_X) \cdot (Y - \mu_Y)]. \tag{1}$$

  - If $\sigma_{XY} > 0$ then X and Y have a positive *linear* association.
  - If $\sigma_{XY} < 0$ then X and Y have a negative *linear* association.
  - If $\sigma_{XY} = 0$ then X and Y have no *linear* association.

Correlation

- The **correlation** is a measure of the direction and strength of the *linear association* between two random variables.
  - The correlation coefficient takes on values between -1 and 1.

- Correlation between two (population) random variables X and Y:

$$\rho = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}. \tag{2}$$

- Where σ_X and σ_Y are the standard deviations of X and Y.
  - If $\rho = -1$ then there is a perfect (negative) linear relationship between X and Y.
  - If $\rho = 1$ then there is a perfect (positive) linear relationship between X and Y.
  - If ρ is close to 0 then there is a weak linear relationship between X and Y.
- The strength of the linear association increases as ρ moves away from 0.

Example 1

- Determine the nature of the linear association between the two variables in each of the 5 example plots in the *Linear Associations Examples* R Markdown file.

Sample Covariance

- Often we do not have all of the observations in populations and we need to estimate parameters.
- **Sample covariance** estimates the actual covariance between two random variables:

$$S_{XY} = \frac{1}{n-1} \cdot \sum_{i=1}^{n} (x_i - \bar{x}) \cdot (y_i - \bar{y}). \tag{3}$$

- Where $\bar{x}$ and $\bar{y}$ are the sample means.

Sample Covariance in R

- R function: `cov(x, y, method = "pearson")`
  - `x`: a numeric vector, matrix, or data frame.
  - If `x` is a vector we must give a vector `y`.
  - `method = "pearson"` is the default method.

Covariance Matrix

Variable	X	Y	Z
X	$V[X]$	$Cov(X,Y)$	$Cov(X,Z)$
Y	$Cov(X,Y)$	$V[Y]$	$Cov(Y,Z)$
Z	$Cov(X,Z)$	$Cov(Y,Z)$	$V[Z]$

Example 2

- Using R, calculate the sample covariance between each of the pairs of variables used in Example 1
- What do these values imply?

Sample Correlation

- Often we do not have all of the observations in populations and we need to estimate parameters.
- **Sample correlation** estimates the actual correlation between two random variables:

$$\rho = \frac{S_{XY}}{S_X \cdot S_Y} = \frac{S_{XY}}{\sqrt{S_X^2 \cdot S_Y^2}}. \tag{4}$$

- Where S_{XY}, S_X, and S_Y are the sample covariance and standard deviations.

Sample Correlation in R

- R function: `cor(x, y, method = "pearson")`
  - `x`: a numeric vector, matrix, or data frame.
  - If `x` is a vector we must give a vector `y`.
  - `method = "pearson"` is the default method.

- `method = "kendall"` to measure the ordinal association between two measured quantities.
- `method = "spearman"` (rank correlation) to assess monotonic relationships between two measured quantities.

Correlation Matrix

Variable	X	Y	Z
X	$\rho_{XX} = 1$	ρ_{XY}	ρ_{XZ}
Y	ρ_{XY}	$\rho_{YY} = 1$	ρ_{YZ}
Z	ρ_{XZ}	ρ_{YZ}	$\rho_{ZZ} = 1$

Example 3

- Using R, calculate the sample correlation between each of the pairs of variables used in Example 1
- What do these values imply?

Properties of Sample Correlation

- Values near 0 indicate a weak linear relationship.
- Linear relationship strength increases as ρ moves towards 1 or -1.
- If ρ is close to 1 or -1 then the scatterplot will be close to a straight line.

Effect of Outliers

- An *outlier* is a data point that differs significantly from other observations.
- As the measures are based on differences from mean values, **outliers can have a large impact on the estimates of linear association.**
- We have covered methods to identify possible outliers (boxplots).
- Being aware of the impacts out potential outliers can help you make better statistical decisions.

Example 4

- Load the *Outliers.csv* data into your workspace to complete the following:
  1. Generate a boxplot for each of the four variables.
  2. Generate a scatterplot for each of the pairs of variables (Y1 vs X1 and Y2 vs X2).
     - Include the correlation coefficient in your plots.
  3. Generate a scatterplot for each of the pairs of variables (Y1 vs X1 and Y2 vs X2) **without** the outliers.
     - Include the correlation coefficient in your plots.
- What did you notice?

Example 5

- Using the *Football24.csv* data complete the following:
  1. Create the functions under the *Required Functions* heading.
  2. Create a pairs plot of all of the numeric variables.
  3. Create a pairs plot of all the numeric variables and include a LOWESS curve, the correlation coefficients, and the histograms.
- Does there appear to be any linear relationships between the variables?

Next Topic

- **Linear Regression I**
- Regression formulation.
- Regression estimation.
- Assumptions and diagnostics.

Exercise 1

- Load the *Cars93* dataset from the *MASS* R package and follow the steps to perform an ANOVA test to determine if there are differences in the *Horsepower* of the three *Cylinders*:

  1. Validate the normality assumption of the three groups.
  2. Validate the equal variance assumption of the three groups.
  3. Perform an appropriate ANOVA test and comment on your results.

Exercise 2

- Load the *Cars93* dataset from the *MASS* R package and follow the steps to perform an ANOVA test to determine if there are differences in the *Price* of the three *Type*:

  1. Validate the normality assumption of the three groups.
  2. Validate the equal variance assumption of the three groups.
  3. Perform an appropriate ANOVA test and comment on your results.

Exercise 3

- Load the *Cars93* dataset from the *MASS* R package and use the Chi-squared tests to test the following:

  1. If the horsepower *(Horsepower)* is independent of the vehicle type (*Type*).
  2. If the highway miles per gallon (*MPG.highway*) is independent of the number of the number of airbags *(AirBags)*
  3. If the price (*Price*) is independent of the number of airbags *(AirBags)*.

References & Resources

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